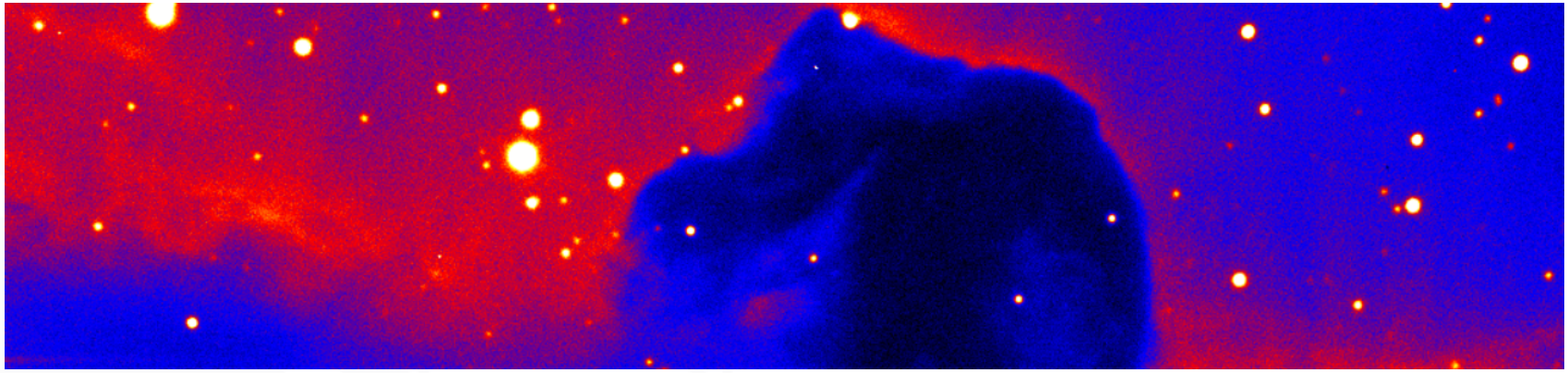


Introduction to FITS images, DS9 and Astropy

M1 AstroCCP HAP 703P
Astrophysique Observationnelle 1

Julien Morin

23rd September 2026

FACULTÉ DES SCIENCES DE MONTPELLIER
LUPM LABORATOIRE MONTPELLIER UNIVERS & PARTICULES

Outline

1. Loging in on the B12 machines
2. The Flexible Image Transport System (FITS)
3. Viewing FITS files with SAO Image DS9
4. Viewing/Analysing data with Python/Astropy
5. Bibliographic resources

Outline

1 Loging in on the B12 machines

2 The Flexible Image Transport System (FITS)

3 Viewing FITS files with SAO Image DS9

4 Viewing/Analysing data with Python/Astropy

5 Bibliographic resources

Loging in on the B12 machines

- Select Skolelinux OS
- Login: nom-p
- Password: student number
- Updating the password: `https://10.10.16.17/gosa/`
- HAP703P moodle: `https://moodle.umontpellier.fr/course/view.php?id=35142`

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The Flexible Image Transport System (FITS)

- Open standard managed by the IAU FITS Working Group
- Unified standard for astronomical data exchange used by ESA, ESO, NASA...
  - 1979: initial development/use
  - 1982: v1.0 standard
  - 2008: v3.0 standard
- FITS format embeds
  - Data (images, spectra, data cubes) in binary format
  - Header in ASCII format
    - Data description: date, # of pixels, telescope, instrument, exposure time...
    - Processing: original images, parameters used...
    - Calibration: WCS calibration, wavelength calibration
- Set of libraries, tools to read/edit FITS
  - CFITSIO C-library + wrappers such as `astropy.io.fits` in python
  - Viewers/Editors: SAO Image DS9, Fv, The Gimp, FITS Liberator
  - Processing softwares: IRAF, AstroImageJ, Audela, IRIS, Siril

FITS file structure

- Header/Data Units (HDUs)
  - Header Unit + Data Unit
- FITS contains:
  - 1 Primary HDU w/ 1-999 dimension array of int/float
  - + optional extension HDUs (i.e. one file may contain **several** images)
    - 'IMAGE': similar to primary array
    - 'TABLE': tabular information in ASCII format
    - 'BINTABLE': tabular information in binary format
- Header Unit
  - `KEYNAME = value / comment string`
- Data Unit: data types for primary array / image extension
  - integer 8/16/32 bits
  - float 32/64 bits (single/double precision)

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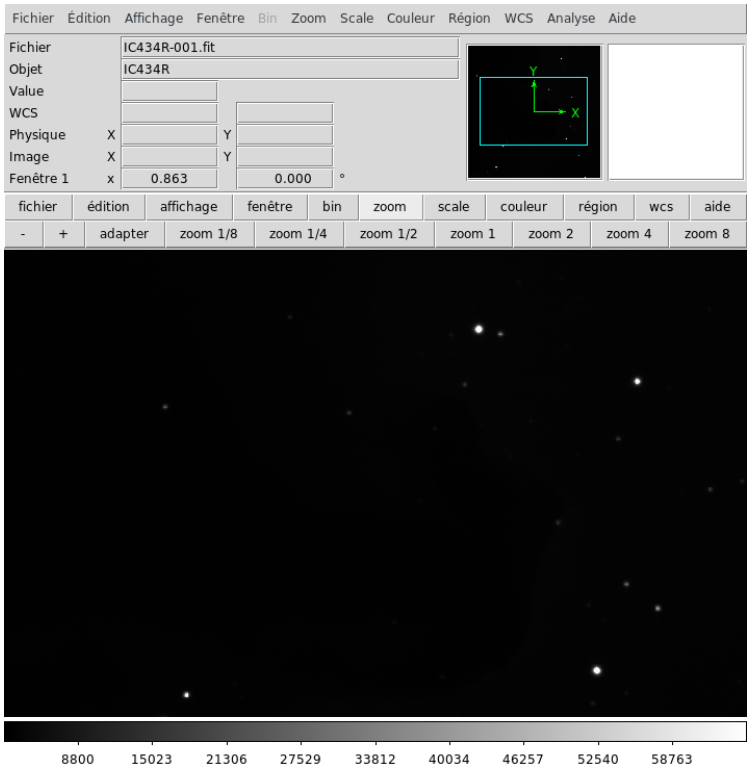
4 Viewing/Analysing data with Python/Astropy

5 Bibliographic resources

Viewing FITS files with SAO Image DS9

- Retrieve tarball containing FITS files on moodle
  - `$ tar -xvzf intro-iraf-files-2018.tgz`
- Header Inspection
  - `File` ➔ `Display Header`
- Image settings
  - `Scale` ➔ `linear/log/zscale`

  ➥ similar to gamma settings
  - `Color` ➔ `b`
- Inspect pixel values
- Perform cuts/averages



DS9 - opening an image

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Fichier Édition Police

```
SIMPLE  =                    T
BITPIX  =                   16 /8 unsigned int, 16 & 32 int, -32 & -64 real
NAXIS   =                    2 /number of axes
NAXIS1  =                 1024 /fastest changing axis
NAXIS2  =                 1024 /next to fastest changing axis
BSCALE  =   1.0000000000000000 /physical = BZERO + BSCALE*array_value
BZERO   =   32768.000000000000 /physical = BZERO + BSCALE*array_value
DATE-OBS= '2014-11-19T02:04:52' /YYYY-MM-DDThh:mm:ss observation start, UT
EXPTIME =   100.00000000000000 /Exposure time in seconds
EXPOSURE=   100.00000000000000 /Exposure time in seconds
SET-TEMP=  -80.000000000000000 /CCD temperature setpoint in C
CCD-TEMP=  -79.896003723144531 /CCD temperature at start of exposure in C
XPIXSZ  =   27.000000000000000 /Pixel Width in microns (after binning)
YPIXSZ  =   27.000000000000000 /Pixel Height in microns (after binning)
XBINNING=                    2 /Binning factor in width
YBINNING=                    2 /Binning factor in height
XORGSUBF=                    0 /Subframe X position in binned pixels
YORGSUBF=                    0 /Subframe Y position in binned pixels
FILTER  = 'R Cousins' /          Filter used when taking image
IMAGETYP= 'Light Frame' /        Type of image
FOCALLEN=   7200.0000000000000 /Focal length of telescope in mm
APTDIA  =   1200.0000000000000 /Aperture diameter of telescope in mm
APTAREA =   1130973.3867645264 /Aperture area of telescope in mm^2
SBSTDVER= 'SBFITSEXT Version 1.0' /Version of SBFITSEXT standard in effect
SWCREATE= 'MaxIm DL Version 6.05 151004 07EE5' /Name of software
SWSERIAL= '07EE5-9M9UR-YVXE4-52N3R-44P77-P6' /Software serial number
SITELAT = '00 00 00' /           Latitude of the imaging location
SITELONG= '00 00 00' /           Longitude of the imaging location
JD      =   2456980.5867129629 /Julian Date at start of exposure
OBJECT  = 'IC434R  '
TELESCOP= '        ' /           telescope used to acquire this image
INSTRUME= 'Andor Tech'
OBSERVER= '        '
NOTES   = '        '
FLIPSTAT= 'Mirror/Rotate 90 CCW'
SWOWNER = 'LE VAN SUU Auguste' /Licensed owner of software
END
```

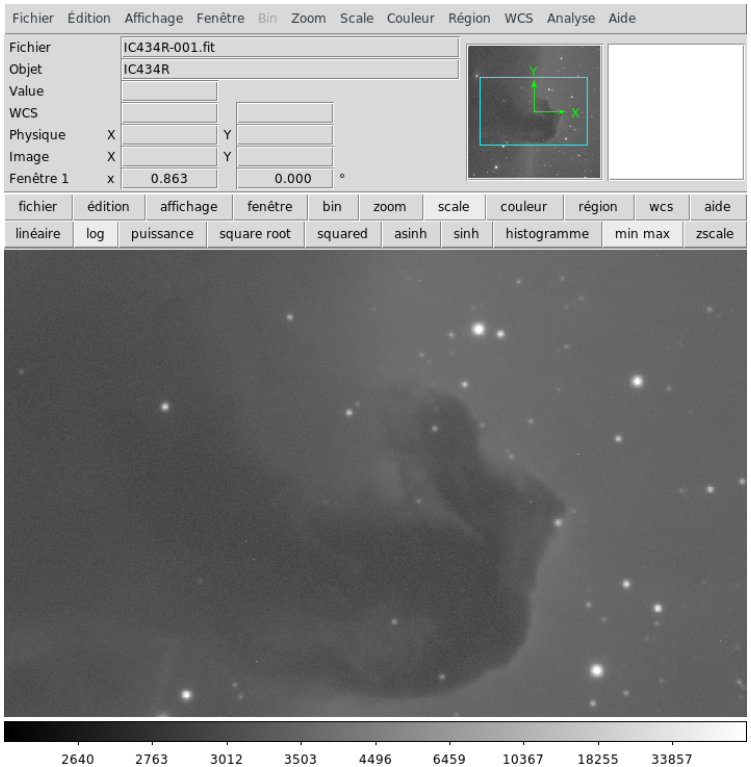
DS9 - displaying image header

Viewing FITS files with SAO Image DS9

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➥ similar to gamma settings

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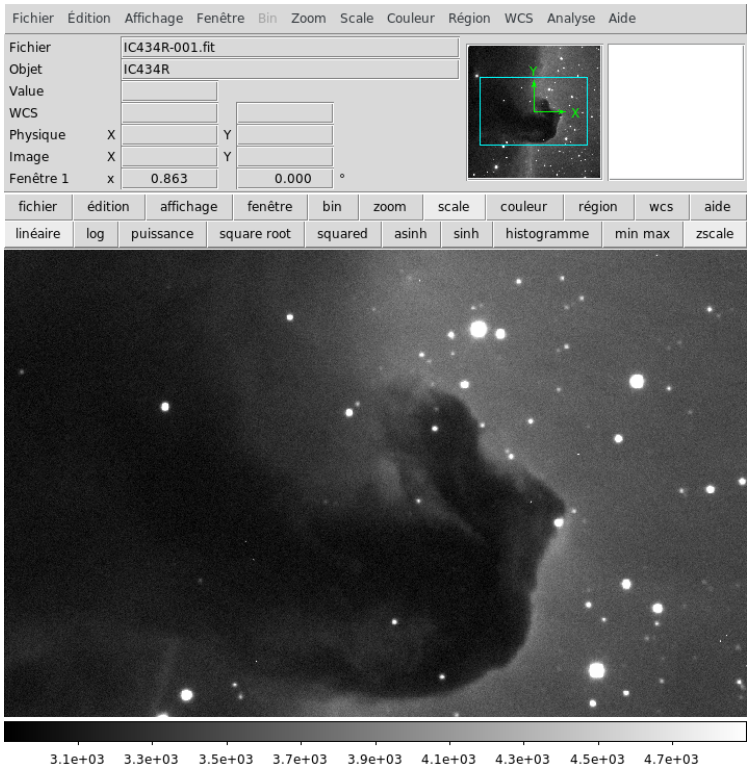


DS9 - setting log scale

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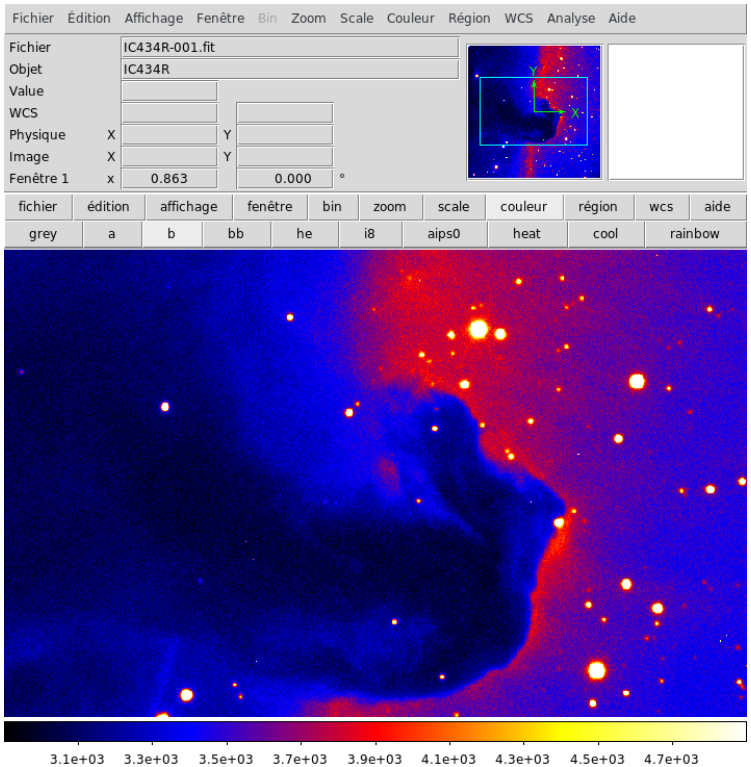


DS9 - setting linear scale + zscale

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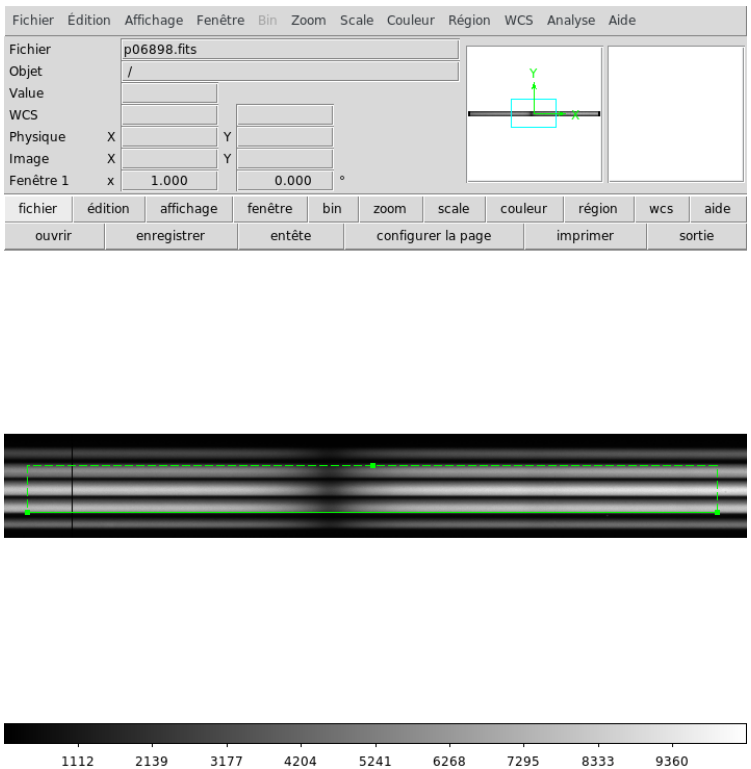


DS9 - setting false color scheme

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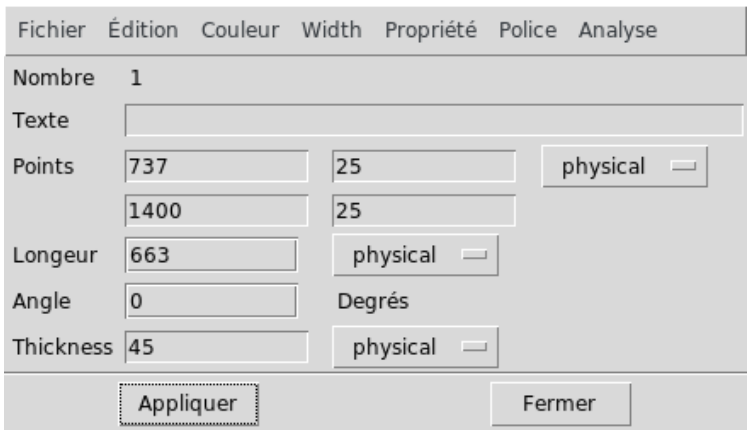
DS9 - "projection"

Viewing FITS files with SAO Image DS9

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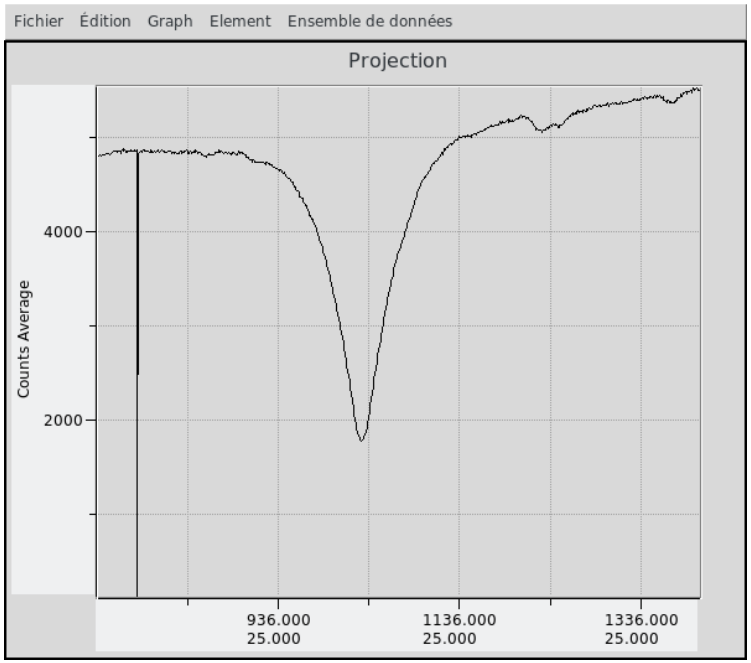
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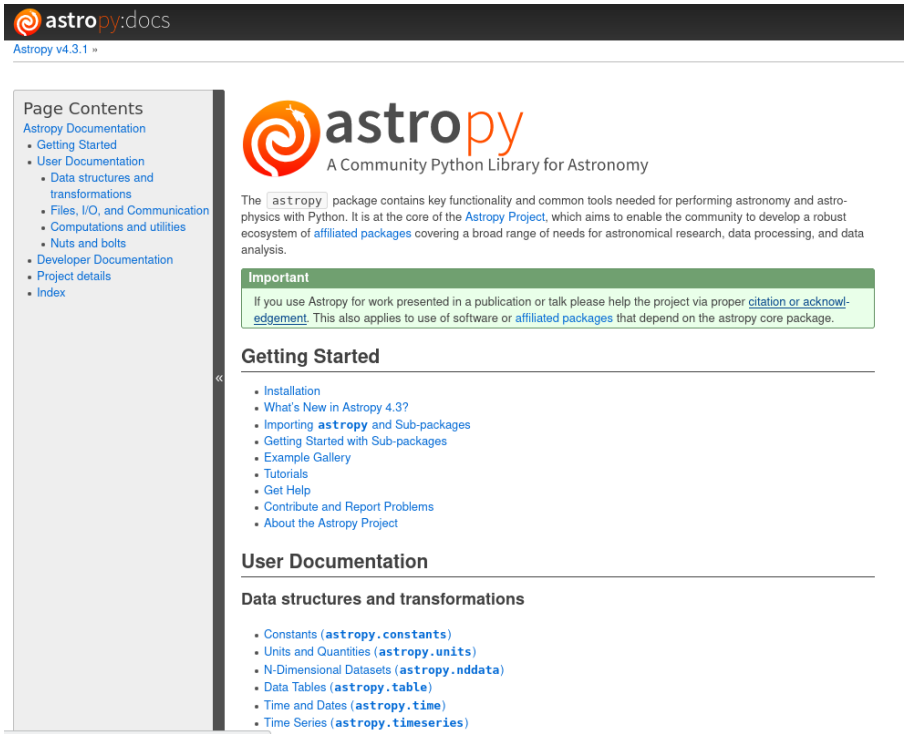
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Getting started with Astropy

- open a terminal
- start an ipython session
- import astropy and load image

```
--> import astropy as ap
--> import astropy.io.fits as pf
--> hdul =
    pf.open('IC434R-001.fit')
```



Astropy documentation page

Displaying FITS content with Astropy

- Display a FITS content: HDUs

--> `hdul.info()`
  - display the HDUs contained in `IC434R-001.fit`

- Display information for `NICMOSn4hk12010_mos.fits`
  - same for a file w/ multiple HDUs

Displaying FITS content with Astropy: header

- Display a FITS header

--> `hdul[0].header`
  - the full header of `IC434R-001.fit`

--> `pf.getheader('IC434R-001.fit')`
  - same w/o explictly loading HDU list beforehand

- Get a specific keyword value/comment

--> `hdul[0].header['EXPTIME']`
--> `hdul[0].header.comments['EXPTIME']`

- List available keywords

--> `hdul[0].header.keys()`
--> `hdul[0].header.comments['EXPTIME']`

- TODO : write a python script that takes a FITS file name as argument and writes down the following keywords: DATE-OBS, OBJECT, NAXIS1, NAXIS2, EXPOSURE

Displaying FITS content with Astropy: data

- Load FITS image data

```
--> data = hdul[0].data
--> data = hdul['PRIMARY'].data
```

  - Call by position or label

- Display a FITS image

```
--> import matplotlib.pyplot as plt
--> im = plt.imshow(data, origin='lower')
--> plt.show()
```

  - option `origin='lower'` results in same orientation as ds9

```
--> from astropy.visualization import astropy_mpl_style
--> plt.style.use(astropy_mpl_style)
```

  - optimize plot parameters for image visualization

FITS with Astropy: data visualization

- Scale/normalize values

```
--> from astropy.visualization import simple_norm
--> norm = simple_norm(data, 'sqrt')
--> plt.imshow(data, origin='lower', norm=norm)
```

  - scale/stretch options available : 'linear', 'sqrt', 'power', 'log', 'asinh'
  - set levels w/ parameters `min_cut`/`max_cut` or `min_percent`/`max_percent`
  - See `simple_norm` documentation

- Set levels only

```
--> from astropy.visualization import MinMaxInterval
--> interval = MinMaxInterval()
--> z1, z2 = interval.get_limits(data)
--> im = plt.imshow(data, origin='lower', vmin=z1, vmax=z2)
```

  - Other options available : `ManualInterval`, `PercentileInterval`, `AsymmetricPercentileInterval`, and `ZScaleInterval`

- For more details see Astropy documentation:

`https://docs.astropy.org/en/stable/visualization/normalization.html`

FITS with Astropy: data statistics

- Compute histogram

```
--> plt.hist(data.ravel(), bins=100, log=True)
--> plt.show()
```

  - NB: `data.ravel()` converts the image into a 1-D array
  - User needs to determine the optimal number of bins

- Use `astropy.visualisation` implementation

```
--> from astropy.visualization import hist
--> hist(data.ravel(), bins='scott', log=True)
```

  - Methods `'scott'` and `'freedman'` provide automatic determination of # bins

- TODO : implement an imstat script that displays : numbers of pixels, min, max, mean, std deviation, mode and robust std deviation based on MAD w/ `astropy.stats.mad_std`

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Bibliographic resources

- *A Beginner's Guide to Working with Astronomical Data*, Markus Pössel, arXiv:1905.13189
- *The FITS Support Office at NASA/GSFC*
  `https://fits.gsfc.nasa.gov/`
- *Documentation Astropy*
  `https://docs.astropy.org/en/stable/index.html`
- *"Handbook of CCD Astronomy"*, S.B. Howell,
  Cambridge Observing Handbooks for Research Astronomers (Book 9),
  Cambridge University Press, 2006
  ISBN-13: 978-0521617628
  - Appendix C: "Some basics of image displays and color images"